

# fixed

## Fixed

**Revision:** 37 **Last modified:** 2026-06-02T00:16:14Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db\_to\_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: **199**.

| HXC      | Type    | Pri | Title                                                                     |
|----------|---------|-----|---------------------------------------------------------------------------|
| HXC-001  | Task    | P1  |                                                                           |
| HXC-002  | Task    | P1  |                                                                           |
| HXC-003  | Task    | P1  |                                                                           |
| HXC-004  | Task    | P1  |                                                                           |
| HXC-005  | Task    | P1  |                                                                           |
| HXC-006  | Task    | P1  |                                                                           |
| HXC-007  | Task    | P1  |                                                                           |
| HXC-008  | Task    | P1  |                                                                           |
| HXC-009  | Task    | P1  |                                                                           |
| HXC-010  | Task    | P1  |                                                                           |
| HXC-011  | Task    | P1  |                                                                           |
| HXC-012  | Task    | P1  |                                                                           |
| HXC-013  | Task    | P1  |                                                                           |
| HXC-014  | Task    | P1  |                                                                           |
| HXC-015  | Task    | P1  |                                                                           |
| HXC-1001 | Task    | P0  | Stand up embedded-etcd integration harness for pkg/etcd tests             |
| HXC-1002 | Feature | P0  | Prove etcd Put/Get round-trip and missing-key semantics against real etcd |
| HXC-1003 | Feature | P1  | Prove etcd GetPrefix returns exactly matching multi-key set               |
| HXC-1004 | Feature | P1  | Prove etcd Delete/DeletePrefix remove keys (sink-side absence)            |
| HXC-1005 | Feature | P0  | Prove etcd Watch fires correct event and closes on ctx cancel             |
| HXC-1006 | Feature | P1  | Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior          |
| HXC-1007 | Feature | P0  | Prove etcd distributed Lock mutual exclusion (highest-risk path)          |
| HXC-1008 | Task    | P2  | Add mutation-paired etcd New() default-timeout injection test             |
| HXC-1009 | Task    | P2  | Prove etcd error-wrapping on dead endpoint                                |

| HXC      | Type    | Pri | Title                                                                       |
|----------|---------|-----|-----------------------------------------------------------------------------|
| HXC-1010 | Feature | P0  | Replace pkg/infra map-echo orchestrator with real service Boot via tests    |
| HXC-1011 | Feature | P0  | Make pkg/infra Health probe real services and report false on failure       |
| HXC-1012 | Feature | P1  | Make pkg/infra Logs capture real sink-side output honoring tail/since/till  |
| HXC-1013 | Feature | P1  | Make pkg/infra Scale create real replicas verified via runtime query        |
| HXC-1014 | Feature | P1  | Make pkg/infra VMSpawn/VMSSH establish a real session end-to-end            |
| HXC-1015 | Feature | P2  | Make pkg/infra VMSimulateFailure/VMSimulatePartition assert observability   |
| HXC-1016 | Feature | P0  | Add EtcdLocker integration tests against real etcd (block/acquire/lease)    |
| HXC-1017 | Bug     | P0  | Rewrite TestMemoryLockerConcurrent to detect a broken lock without test     |
| HXC-1018 | Task    | P2  | Add explicit block-then-acquire ordering test for MemoryLocker              |
| HXC-1019 | Bug     | P2  | Characterize or fix MemoryLocker.Lock context-cancellation goroutine        |
| HXC-1020 | Feature | P0  | Wire a real Builder behind pkg/build Service and prove image is pullable    |
| HXC-1021 | Bug     | P1  | Drive pkg/build StateFailed via a genuine build failure not the 'fail' sent |
| HXC-1022 | Bug     | P1  | Make pkg/build List/Concurrent tests Start() the service and assert termi   |
| HXC-1023 | Task    | P2  | Replace time.Sleep polling in pkg/build with completion sync and add s      |
| HXC-1024 | Bug     | P2  | Enforce or document pkg/build content-addressable cache digest integri      |
| HXC-1025 | Task    | P2  | Cover pkg/build cancel-mid-flight branch sets StateCancelled                |
| HXC-1026 | Feature | P0  | Add real mTLS handshake test for pkg/security (valid client succeeds, n     |
| HXC-1027 | Bug     | P1  | Assert MinVersion==TLS13 on the client TLS path in pkg/security             |
| HXC-1028 | Feature | P0  | Locate/build production Vault client and add dev-Vault integration test (   |
| HXC-1029 | Task    | P3  | Replace pkg/security non-nil-only TestNewTLSConfigBuilder with mea          |
| HXC-1030 | Bug     | P1  | Verify pkg/hxcregistry item_history audit-sink rows after Create/Update     |
| HXC-1031 | Bug     | P2  | Assert exact and stable ComputeHeadingHash value in pkg/hxcregistry         |
| HXC-1032 | Task    | P2  | Add pkg/hxcregistry failure-path tests (dup PK, missing-id Get/Update)      |
| HXC-1033 | Bug     | P2  | Make pkg/hxcregistry concurrent test do concurrent writes (or rename)       |
| HXC-1034 | Task    | P3  | Prove pkg/hxcregistry timestamp round-trip and DB CHECK-constraint          |
| HXC-1035 | Bug     | P0  | Wire pkg/wasm WasmPlugin.Execute to real WASM output and prove tr           |
| HXC-1036 | Task    | P1  | Add pkg/wasm negative tests for missing execute export and corrupt mo       |
| HXC-1037 | Bug     | P2  | Prove pkg/wasm init/shutdown exports actually run via memory side-eff       |
| HXC-1038 | Task    | P3  | Cover pkg/wasm Host.Call ErrUnexpectedResult branch                         |
| HXC-1039 | Bug     | P1  | Replace self-defeating pkg/semaphore TestSemaphore_ReleaseTooMany           |
| HXC-1040 | Feature | P1  | Add pkg/semaphore M>>N concurrency invariant test under -race               |
| HXC-1041 | Task    | P3  | Add pkg/semaphore New(0) barrier and exact-capacity boundary tests          |
| HXC-1042 | Bug     | P1  | Convert pkg/storage error tests to errors.Is on exported sentinels          |
| HXC-1043 | Bug     | P2  | Reconcile and test pkg/storage FileStore vs MemoryStore copy semanti        |
| HXC-1044 | Task    | P2  | Add pkg/storage overwrite, missing-delete, and List edge tests for both s   |
| HXC-1045 | Feature | P1  | Add pkg/storage -race concurrent Put/Get/Delete test to prove RWMutex       |
| HXC-1046 | Bug     | P1  | Gate -race in CI for pkg/metrics concurrency mutation tests                 |
| HXC-1047 | Bug     | P2  | Complete pkg/metrics Prometheus exposition assertions (+Inf, _sum, or       |

| HXC      | Type    | Pri | Title                                                                            |
|----------|---------|-----|----------------------------------------------------------------------------------|
| HXC-1048 | Task    | P2  | Add pkg/metrics duplicate-registration and histogram edge tests                  |
| HXC-1049 | Task    | P3  | Add pkg/metrics /metrics integration test via httptest.Server                    |
| HXC-1050 | Bug     | P1  | Test pkg/grpcutil LoggingStreamInterceptor recv/send counters                    |
| HXC-1051 | Bug     | P2  | Strengthen pkg/grpcutil log assertions to dynamic behavior-dependent fi          |
| HXC-1052 | Task    | P2  | Cover pkg/grpcutil status.FromError code extraction for handler errors           |
| HXC-1053 | Bug     | P1  | Add pkg/context TestDetach_PreservesValues for the value-retention gu            |
| HXC-1054 | Bug     | P2  | Strengthen pkg/context WithTimeout deadline and expiry assertions                |
| HXC-1055 | Bug     | P1  | Give pkg/workerpool TestPool_NilWorkSafe_Mutation a real sink                    |
| HXC-1056 | Feature | P1  | Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recover, tests       |
| HXC-1057 | Bug     | P1  | Make pkg/errors concurrency test honest via locked GetFields path + -ra          |
| HXC-1058 | Task    | P2  | Pin pkg/errors Code enum exact values, distinctness, and WithFields nil          |
| HXC-1059 | Bug     | P2  | Replace tautological pkg/backoff cap-mutation test and pin Default()             |
| HXC-1060 | Bug     | P2  | Assert pkg/classads parser AST Op per operator and paren precedence              |
| HXC-1061 | Task    | P3  | Add pkg/classads eval failure-path and regexp non-match tests                    |
| HXC-1062 | Bug     | P2  | Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is c             |
| HXC-1063 | Task    | P3  | Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBI             |
| HXC-1064 | Bug     | P1  | Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug             |
| HXC-1065 | Task    | P3  | Strengthen pkg/log Warn test, add Debug-suppression, remove dead cap scaffolding |
| HXC-1066 | Task    | P3  | Add pkg/lru boundary, cold-miss, and struct-value round-trip tests               |
| HXC-1067 | Feature | P2  | Add pkg/pubsub -race concurrency test and explicit drop-boundary asse            |
| HXC-1068 | Bug     | P2  | Prove pkg/ratelimit bucket starts full and add Limiter conformance + -ra         |
| HXC-1069 | Bug     | P1  | Bring pkg/retry DoWithResult to behavioral parity with Do                        |
| HXC-1070 | Task    | P2  | Add pkg/retry transient-recovery, jitter-bound, and computeDelay table           |
| HXC-1071 | Task    | P3  | Strengthen pkg/serde MustMarshal exact-content and add Unmarshal fa              |
| HXC-1072 | Bug     | P2  | Pin pkg/validator error contract via errors.Is and remove/wire unused se         |
| HXC-1073 | Task    | P3  | Add pkg/validator uint min/max, IsValidID empty, oneof/email boundar             |
| HXC-1074 | Task    | P0  | Establish fleet-wide -race CI gate across concurrency-sensitive packages         |
| HXC-1075 | Feature | P0  | Implement NATS/JetStream client wrapper (pkg/events)                             |
| HXC-1076 | Feature | P0  | Implement etcd client wrapper (pkg/etcd)                                         |
| HXC-1077 | Feature | P0  | Implement etcd discovery backend (pkg/discovery)                                 |
| HXC-1078 | Feature | P0  | Implement real Kafka producer/consumer (pkg/pubsub)                              |
| HXC-1079 | Feature | P0  | Implement cgroups v2 + /proc resource reader (pkg/resources)                     |
| HXC-1080 | Feature | P0  | Implement Omega-model scheduler core (pkg/scheduler, internal/schedu             |
| HXC-1081 | Feature | P0  | Implement optimistic-concurrency versioning for scheduler pool                   |
| HXC-1082 | Feature | P1  | Implement HTCondor ClassAds matching in scheduler (pkg/classads)                 |
| HXC-1083 | Feature | P1  | Implement gang scheduling and preemption in scheduler                            |
| HXC-1084 | Feature | P0  | Upgrade tmux backend to control-mode (-CC) with streaming I/O (pkg/              |
| HXC-1085 | Feature | P0  | Implement CRDT session state and lifecycle (create/attach/migrate)               |

| HXC      | Type    | Pri | Title                                                                               |
|----------|---------|-----|-------------------------------------------------------------------------------------|
| HXC-1086 | Feature | P0  | Implement real Node gRPC handlers serving helixv1 Node API (internal/node)          |
| HXC-1087 | Feature | P0  | Make node registry etcd-backed and cross-process shared (internal/node)             |
| HXC-1088 | Feature | P0  | Implement SWIM gossip membership and failure detection (pkg/swim)                   |
| HXC-1089 | Feature | P0  | Implement WireGuard mesh config-gen and key rotation (pkg/wireguard)                |
| HXC-1090 | Feature | P1  | Implement WireGuard NAT traversal / STUN hole-punch (pkg/wireguard)                 |
| HXC-1091 | Feature | P0  | Wire API gateway reverse proxy with prefix routing and /health (internal/gateway)   |
| HXC-1092 | Feature | P0  | Add JWT/RBAC auth middleware to API gateway (internal/gateway)                      |
| HXC-1093 | Feature | P0  | Implement leader election with fencing tokens (pkg/leader)                          |
| HXC-1094 | Feature | P0  | Implement distributed lock primitives over etcd (pkg/lock)                          |
| HXC-1095 | Feature | P1  | Implement Postgres-backed internal artifact registry (pkg/hxcregistry)              |
| HXC-1096 | Feature | P1  | Implement GPU resource probe with /proc parser (internal/gpu)                       |
| HXC-1097 | Feature | P1  | Define vendor-agnostic GPUBackend interface and registry (internal/gpu)             |
| HXC-1098 | Feature | P2  | Implement GPU sharing modes (MPS, time-slice, MIG, exclusive)                       |
| HXC-1099 | Feature | P1  | Implement Build Service orchestrator and worker pool (internal/build)               |
| HXC-1101 | Feature | P0  | Implement e2ee transport record protocol (security/pkg/e2ee)                        |
| HXC-1102 | Feature | P1  | Implement software-rooted attestation (security/pkg/attestation)                    |
| HXC-1103 | Feature | P0  | Implement RBAC scopes and identity bindings (internal/security)                     |
| HXC-1104 | Feature | P1  | Implement SPIFFE/SPIRE SVID issuance and mTLS identity (internal/security)          |
| HXC-1105 | Feature | P0  | Implement htmux CLI raw-mode terminal client (cmd/htmux)                            |
| HXC-1106 | Feature | P0  | Implement session I/O PTY-over-WebSocket forwarding (cmd/helix-session)             |
| HXC-1107 | Feature | P0  | Implement WebSocket envelope and MessagePack framing (pkg/websocket)                |
| HXC-1108 | Feature | P1  | Implement HelixQA live-service Challenge runner (cmd/helix-test)                    |
| HXC-1109 | Feature | P1  | Implement deterministic-sim, chaos, and device-sim test harness (cmd/helix-test)    |
| HXC-1110 | Task    | P1  | Wire Prometheus /metrics endpoint uniformly across all services (cmd/helix-test)    |
| HXC-1111 | Task    | P1  | Wire Jaeger/OTel trace-ID propagation across services (pkg/tracing)                 |
| HXC-1112 | Task    | P1  | Wire pkg/health gRPC health protocol on every service (pkg/health, cmd/helix-test)  |
| HXC-1114 | Feature | P0  | Provision PostgreSQL primary schema with 15 tables, indexes, and triggers           |
| HXC-1115 | Task    | P0  | Define etcd key namespace and key-builder constants (pkg/etcd)                      |
| HXC-1116 | Feature | P1  | Define Redis cache, routing, and rate-limit key structure (pkg/storage)             |
| HXC-1118 | Feature | P1  | Implement REST API surface on API Gateway with OpenAPI 3.0                          |
| HXC-1120 | Task    | P1  | Configure Kafka topics with partitions/replication/retention                        |
| HXC-1121 | Task    | P0  | Configure NATS JetStream streams (HELIX_NODES/SESSIONS/SCHEDULE/HEALTH/ALERTS)      |
| HXC-1122 | Feature | P2  | Implement Cap'n Proto zero-copy serialization for control-plane (pkg/serialization) |
| HXC-1123 | Feature | P2  | Implement FlatBuffers serialization for GPU compute payloads                        |
| HXC-1124 | Feature | P2  | Implement ZeroMQ message patterns for internal data planes                          |
| HXC-1125 | Feature | P1  | Implement Setup Wizard single-command node onboarding (cmd, BASIS)                  |
| HXC-1127 | Feature | P1  | Implement OPA/WASM Policy Engine with HelixConstitution enforcement                 |
| HXC-1130 | Feature | P2  | Implement Zellij and GNU screen session backends                                    |
| HXC-1131 | Feature | P2  | Implement Backup Service (etcd snapshots, WAL archival, Ceph checkpoints)           |

| HXC      | Type    | Pri | Title                                                                          |
|----------|---------|-----|--------------------------------------------------------------------------------|
| HXC-1132 | Feature | P1  | Implement Metrics Collector node scrape endpoint and GPU metrics aggregation   |
| HXC-1133 | Feature | P1  | Implement token-bucket rate limiter for API gateway (pkg/ratelimit)            |
| HXC-1134 | Feature | P2  | Implement Vault secret injection and rotation for all services                 |
| HXC-1135 | Feature | P0  | Implement two-node cluster formation end-to-end (exit gate)                    |
| HXC-1136 | Feature | P0  | Implement session-create end-to-end path (htmux new exit gate)                 |
| HXC-1137 | Task    | P1  | Benchmark job scheduling decision latency under 100ms (exit gate)              |
| HXC-1138 | Task    | P1  | Enforce >60% pkg/ line coverage and paired mutation tests (Constitutional AI)  |
| HXC-1141 | Feature | P2  | Implement interactive-mode AI CLI agent resource provisioning                  |
| HXC-1142 | Feature | P2  | Implement hybrid mode switching and batch/interactive resource sharing         |
| HXC-1144 | Feature | P2  | Implement WireGuard mesh network policies (segmentation)                       |
| HXC-1148 | Feature | P1  | Replace pkg/wireguard NAT discovery with RFC 5389 STUN client                  |
| HXC-1149 | Task    | P3  | Return typed ErrUnsupported for UPnP/NAT-PMP in pkg/wireguard                  |
| HXC-1150 | Feature | P1  | Add WireGuard key-rotation grace/overlap window with ActiveKeys()              |
| HXC-1151 | Task    | P1  | Add pkg/discovery etcd-backed integration test against a real etcd             |
| HXC-1152 | Feature | P1  | Implement pkg/session container-checkpoint migration strategy (CRDT)           |
| HXC-1156 | Feature | P1  | Implement thermal-aware scheduler plugin (throttle on overheat)                |
| HXC-1157 | Feature | P1  | Implement semi-trusted node output verification (redundant-compute/chaos)      |
| HXC-1160 | Feature | P2  | Improve tmux session backend to full PTY wrap on Attach                        |
| HXC-1162 | Task    | P2  | E2E: multi-node session CRDT convergence                                       |
| HXC-1188 | Feature | P0  | Enforce work-unit resource limits (duration/memory/CPU) on edge devices        |
| HXC-1189 | Feature | P0  | Implement EdgeAwarePlugin scheduler Filter stage                               |
| HXC-1190 | Feature | P1  | Implement EdgeAwarePlugin scheduler Score stage                                |
| HXC-1191 | Feature | P1  | Implement declarative per-tier ScheduleRule engine                             |
| HXC-1195 | Feature | P0  | Implement edge trust-level model (STANDARD/SEMI/EDGE_DONOR)                    |
| HXC-1196 | Feature | P0  | Enforce workload restriction matrix by trust level                             |
| HXC-1198 | Feature | P1  | Implement offline sync protocol with delta compression                         |
| HXC-1199 | Feature | P1  | Implement edge sensor-fusion framework and stream workload type                |
| HXC-1210 | Bug     | P0  | Wire EtcdBackend into the node agent for cluster-wide discovery                |
| HXC-1211 | Bug     | P0  | Make internal/scheduler.StreamJobEvents a real event stream                    |
| HXC-1212 | Feature | P0  | Connect pkg/classads to scheduler Filter/Score stages                          |
| HXC-1213 | Feature | P0  | Add auth + RBAC enforcement to the gateway                                     |
| HXC-1214 | Feature | P1  | Add S3-compatible (minio) backend to pkg/storage                               |
| HXC-1215 | Feature | P1  | Add OTLP/Jaeger export shim for pkg/tracing                                    |
| HXC-1220 | Task    | P2  | Replace custom policy engine with OPA/Rego                                     |
| HXC-1229 | Task    | P0  | Build device profile registry with versioned YAML schema for T1-T8             |
| HXC-1231 | Feature | P1  | Implement device provisioning lifecycle abstraction (Provisioner/Instantiator) |
| HXC-1236 | Feature | P0  | Implement INetwork/HelixNetwork trait with production and simulation           |
| HXC-1237 | Feature | P0  | Wire DST sim transport seam into pkg/swim membership (prod+sim parallel)       |
| HXC-1238 | Feature | P1  | Implement BUGGIFY macro framework with ~25% deterministic fire rate            |

| HXC      | Type     | Pri | Title                                                                                                    |
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| HXC-1239 | Feature  | P1  | Implement 10:1 DST virtual-time compression by fast-forwarding idle p                                    |
| HXC-1240 | Feature  | P1  | Achieve 1,000+ simulated nodes in a single DST process                                                   |
| HXC-1241 | Feature  | P1  | Author DST consensus+gossip workloads using<br>SETUP→EXECUTION→CHECK→METRICS pattern                     |
| HXC-1243 | Feature  | P0  | Implement 8 network fault injectors (latency, loss, corruption, reorder, d<br>partition, DNS, TCP reset) |
| HXC-1244 | Feature  | P0  | Implement 8 node fault injectors (VM crash/restart/pause, CPU/mem/di<br>kill, graceful shutdown)         |
| HXC-1247 | Feature  | P1  | Implement pure-Go in-sim chaos faults toward 25+ (ClockSkew, DiskFi<br>MessageReorder, Byzantine, etc.)  |
| HXC-1248 | Feature  | P0  | Build YAML chaos Scenario Engine with phases, blast radius and abort-                                    |
| HXC-1249 | Feature  | P0  | Implement emergency-stop and auto-recovery with <=2s halt latency                                        |
| HXC-1254 | Feature  | P1  | Implement TestRunner with parallel suite execution and result collection                                 |
| HXC-1255 | Feature  | P1  | Implement session test state machine<br>(IDLE→SETUP→RUNNING→CHAOS_INJECT→VERIFY→RECO                     |
| HXC-1260 | Feature  | P1  | Implement metrics validation against baseline KPI table with severity ga                                 |
| HXC-1261 | Feature  | P1  | Implement Welch's t-test statistical regression detector for HelixQA                                     |
| HXC-1265 | Feature  | P1  | Implement capability-based plugin sandbox with resource limits and auc                                   |
| HXC-1267 | Feature  | P0  | Implement cmd/helix-test CLI with dst/chaos/device/snapshot subcomm                                      |
| HXC-1268 | Feature  | P1  | Build cmd/helix-snapshot standalone CLI reusing pkg/testing/snapshot                                     |
| HXC-1278 | Research | P3  | Add property-based + coverage-guided fuzzing for cluster operations                                      |
| HXC-1301 | Feature  | P1  | Implement pkg/cloudspot PreemptionWatcher with injectable SignalSou<br>hooks                             |
| HXC-1303 | Feature  | P1  | Implement pkg/inference provider-agnostic Backend interface + capabil                                    |
| HXC-1311 | Docs     | P1  | Author complete YAML tier definitions (T1-T15) with min_requirement                                      |
| HXC-902  | Task     | P1  |                                                                                                          |
| HXC-903  | Task     | P1  |                                                                                                          |